Exp 14

29/9/26

Routing at Network Layer:

A) Static Routing protocol configuration

Aim:- To simulate Static Routing Protocol Configuration using CISCO Packet Tracer.

Observation:-

1) Static routing involves manually adding routes to a routes routing table.

2) Routers automatically knows the network directly connected to the interfaces.

3) Routers are manually added for networks that are available through other routers.

4) Both Main and backup routers are configured for a destination

5) Administrative Distance (AD) is used to select the preferred static route.

6) The routing table is checked using 'show ip route static'

7) Connectivity and routing paths are verified using ping and tracert

8) When the main route fails, the backup route can be used for forwarding packets.

Result:- Thus, static routes were successfully configured and verified, including the use of backup route.

B) RIP - Dynamic Routing

Aim:- To simulate and configure RIP (Routing Information protocol) using CISCO Packet Tracer.

Observation:-

1) Three routers are connected to form a network with multiple possible paths.

2) IP address are assigned to the PC's, Fast Ethernet interfaces and Serial interfaces.

3) RIP is enabled using the router's rip command.

4) Directly connected networks are advertised using the network command.

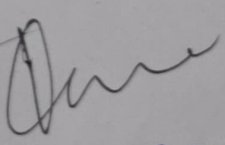
5) RIP automatically exchanges routing information between routers.

6) RIP selects the router based on the lowest hop count

7) The ping command is used to test connectivity between PC0 and PC1.

8) The tracert command is used to verify the path followed by packets.

9) If the selected route fails, RIP automatically chooses another available routes.



Result :- Thus RIP was successfully configured and dynamic routing was verified. The router automatically selected the available route.